

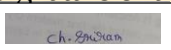
SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science & Engineering
ASSESSMENT TOOL 2 -Technical Presentation

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 2 Technical Presentation
Weightage	30% of CIA	Total Marks	100
CO1 Statement	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
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Section/Batch	Slot C /2025	Date	06-10-2026

Code of Conduct:

I, CH.V.N.V.S SRIRAM , certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



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Technical Presentation
CSA0314 - DATA STRUCTURES

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Problem & Requirement

Problem:

- A navigation system connects different cities and locations.
- Each city may have connections with many other cities.
- Roads have different distances or travel times.
- The system needs to find the **shortest route** between two locations.

Requirements:

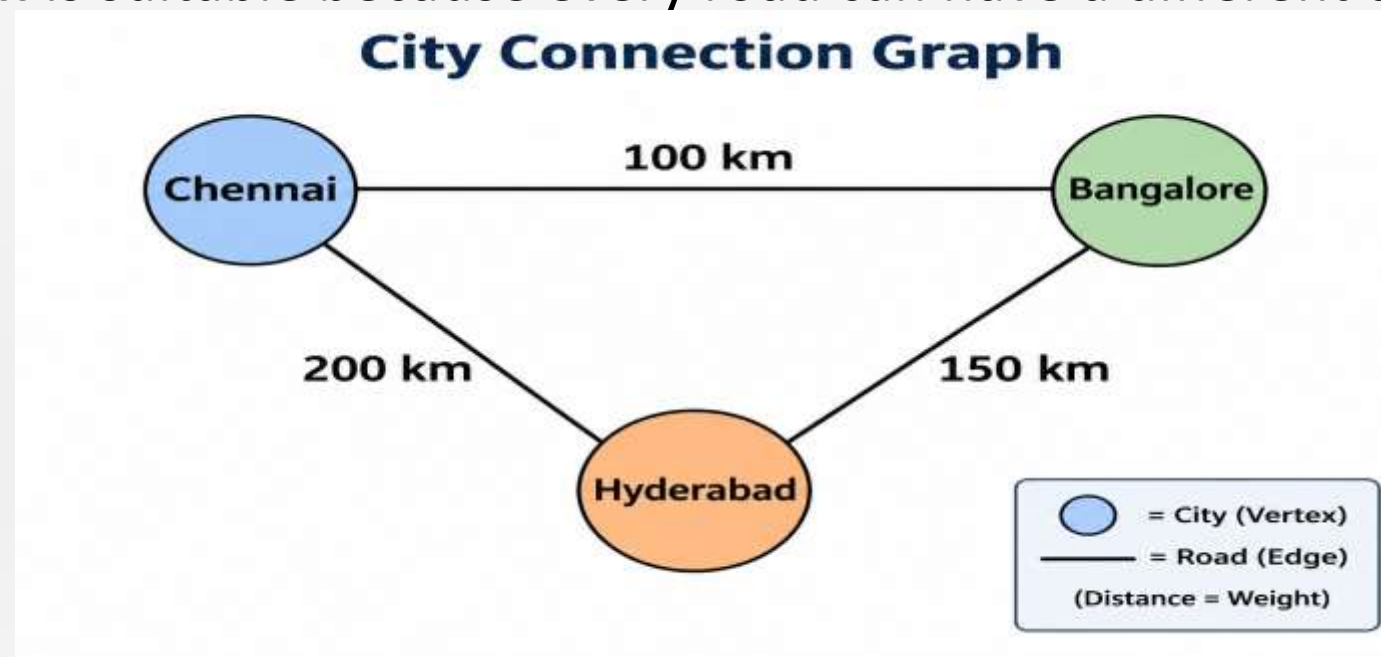
- Store cities.
- Store road connections.
- Store distance/time.
- Find the shortest route.



Suitable Data Structure: Graph

- A **Graph** is a non-linear data structure containing:
- **Vertices (Nodes)** → Cities/locations
- **Edges** → Roads/connections
- **Weights** → Distance or travel time

Weighted Graph is suitable because every road can have a different distance.





Shortest Path Using Dijkstra's Algorithm

Dijkstra's Algorithm can find the shortest route when road weights are non-negative.

Steps:

Select the starting city.

Set its distance as 0.

Set other distances as infinity.

Select the city with the smallest distance.

Update distances of connected cities.

Repeat until the destination is reached.

Example:

$A \rightarrow B = 4 \text{ km}$

$B \rightarrow D = 2 \text{ km}$

Shortest route:

$A \rightarrow B \rightarrow D = 6 \text{ km}$



Why Graph is Appropriate?

Data Structure	Suitability
Array	Less suitable
Linked List	Less suitable
Stack	Not suitable
Queue	Not suitable
Tree	Limited
Graph	Highly suitable



Advantages of Graph

- Represents **city-to-city connections** easily.
- Supports **multiple connections** between locations.
- Stores **distance or travel time** as edge weights.
- Helps find the **shortest route** efficiently.
- Can represent **one-way and two-way roads**.
- Suitable for **large navigation and road networks**.
- Easy to update when **new roads or cities** are added.



Conclusion

- A **weighted graph** is suitable for a navigation system.
- Cities are represented as **vertices**.
- Roads are represented as **edges**.
- Distance or travel time is represented as **weights**.
- **Dijkstra's algorithm** can find the shortest route.
- Graphs are widely useful for GPS, maps, transport and route-planning systems.

